

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 2 – Seminar on Emerging Technologies

Course Code:	CSA1017	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 2 – Seminar on Emerging Technologies
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	MURALI KRISHNAN	Reg. No.:	192524090
Date:		Duration:	60 minutes

Code of Conduct

I NURALI KRISHNAN N (192524090) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Seminar on Emerging Technologies Instructions to Students:

Students must select the topic strictly according to the Serial Number (S.No.) order assigned to them.

S.No.	Seminar Topic
1	Generative AI and Large Language Models (LLMs)
2	AI Agents and Autonomous AI Systems
3	Multimodal Artificial Intelligence
4	Explainable and Trustworthy AI
5	Edge AI and Intelligent Edge Computing
6	Federated Learning and Privacy-Preserving AI
7	Deep Learning for Computer Vision
8	AI-Powered Medical Image Analysis

9	Natural Language Processing with Transformer Models
10	Retrieval-Augmented Generation (RAG)
11	AI in Cybersecurity and Threat Detection
12	Zero Trust Architecture and Modern Cybersecurity
13	Post-Quantum Cryptography
14	Quantum Computing and Quantum Algorithms
15	Quantum Machine Learning
16	Cloud-Native Computing and Microservices
17	Serverless Computing and Function-as-a-Service
18	Edge Computing and Fog Computing
19	Cloud Security and Confidential Computing
20	Digital Twins and Intelligent Simulation
21	Internet of Things (IoT) and Smart Systems
22	Industrial Internet of Things (IIoT)
23	5G/6G Networks and Next-Generation Communication
24	Blockchain Technology and Decentralized Applications
25	Web3 and Decentralized Internet
26	Smart Contracts and Blockchain Applications
27	Augmented Reality (AR) and Virtual Reality (VR)
28	Extended Reality (XR) and Immersive Technologies
29	Robotics and Autonomous Robots
30	Human–Robot Interaction and Collaborative Robots
31	Autonomous Vehicles and Intelligent Transportation Systems
32	Smart Cities and AI-Driven Urban Technologies
33	Metaverse Technologies and Virtual Worlds
34	Brain–Computer Interfaces (BCI)
35	Wearable Computing and Smart Wearable Devices
36	Green Computing and Sustainable IT
37	AI for Software Engineering and Code Generation
38	DevSecOps and Intelligent CI/CD Pipelines
39	Digital Forensics and AI-Based Cybercrime Investigation
40	Privacy-Enhancing Technologies and Data Protection

Seminar Presentation Structure:

1. **Introduction to the Emerging Technology**
2. **Need and Motivation**
3. **Working Principle / Architecture**
4. **Key Components and Technologies**
5. **Applications / Real-World Use Cases**
6. **Advantages and Limitations**
7. **Recent Developments and Trends**
8. **Challenges and Future Scope**
9. **Case Study / Example**
10. **Conclusion and References**

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Technical Content & Subject	Comprehensive and accurate content; demonstrates excellent	Good technical understanding with minor gaps	Basic understanding with limited technical depth	Incomplete content or major technical errors
Knowledge (25%)	understanding of the emerging technology			
Organization & Presentation Structure (15%)	Excellent logical flow with clear introduction, content, conclusion, and effective transitions	Well-structured with minor organizational issues	Adequately structured but some sections lack clarity	Poorly structured and difficult to follow
Examples, Diagrams & Case Study (20%)	Excellent use of relevant diagrams, examples, applications, and real-world case studies	Good use of relevant examples and diagrams	Limited examples/diagrams with basic explanation	Inappropriate or insufficient examples/diagrams

Communication & Presentation Skills (20%)	Very clear, confident, engaging delivery with excellent eye contact, voice modulation, and time management	Clear and confident presentation with minor issues	Understandable but lacks confidence or engagement	Unclear delivery, excessive reading, or poor time management
Question Handling & Overall Performance (20%)	Answers questions accurately and confidently; demonstrates strong overall knowledge	Answers most questions correctly	Answers basic questions with some difficulty	Unable to answer questions or demonstrate adequate understanding

Criteria	Mark
Technical Content & Subject Knowledge (25%)	/5
Organization & Presentation Structure (15%)	/5
Examples, Diagrams & Case Study (20%)	/5
Communication & Presentation Skills (20%)	/5
Question Handling & Overall Performance (20%)	/5
TOTAL	/25

Abstract

A smart microgrid combines distributed energy resources, battery storage, electrical loads, sensors, communication networks and intelligent control. This project proposes an AI-based platform that continuously monitors generation and consumption, forecasts short-term electricity demand, optimizes battery charging and discharging, and balances loads according to operational priorities.

The platform integrates renewable sources such as solar and wind with smart meters and IoT devices. Historical and real-time data are processed by AI models to predict future demand and renewable availability. An optimization and control layer then recommends

or automatically performs suitable actions, such as storing excess renewable energy, supplying priority loads from batteries, shifting flexible loads and reducing dependence on the main grid.

The expected result is improved energy efficiency, better renewable utilization, reduced operating cost, stronger reliability and data-driven planning. A real-time dashboard and periodic reports allow operators to understand energy flow, battery state, load patterns and system performance.

Keywords

Microgrid, Artificial Intelligence, Internet of Things, Smart Meter, Load Balancing, Demand Forecasting, Battery Energy Storage System, Renewable Energy, Energy Management.

1. Introduction to the Emerging Technology

A microgrid is a localized electrical system containing interconnected loads and distributed energy resources. It can coordinate local generation, storage and consumption and may operate with the main grid or, when designed for it, in an independent islanded mode.

Traditional energy management mainly follows predefined rules. These rules can become inefficient when solar and wind generation changes rapidly or when consumer demand varies unexpectedly. AI introduces predictive and adaptive decision-making by learning from historical energy patterns and current operating conditions.

Renewable Generation	Smart Meters	AI Energy Management	Battery Storage	Priority & Flexible Loads
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Figure 1. Conceptual flow of the proposed smart microgrid platform.

2. Need and Motivation

- Renewable generation is intermittent and difficult to predict using only fixed rules.
- Electricity demand changes by time, weather, occupancy and industrial activity.
- Battery storage must be used carefully to avoid unnecessary grid purchases.
- Critical loads require higher reliability than non-essential loads.
- Operators need real-time visibility and clear energy performance reports.

3. Problem Statement

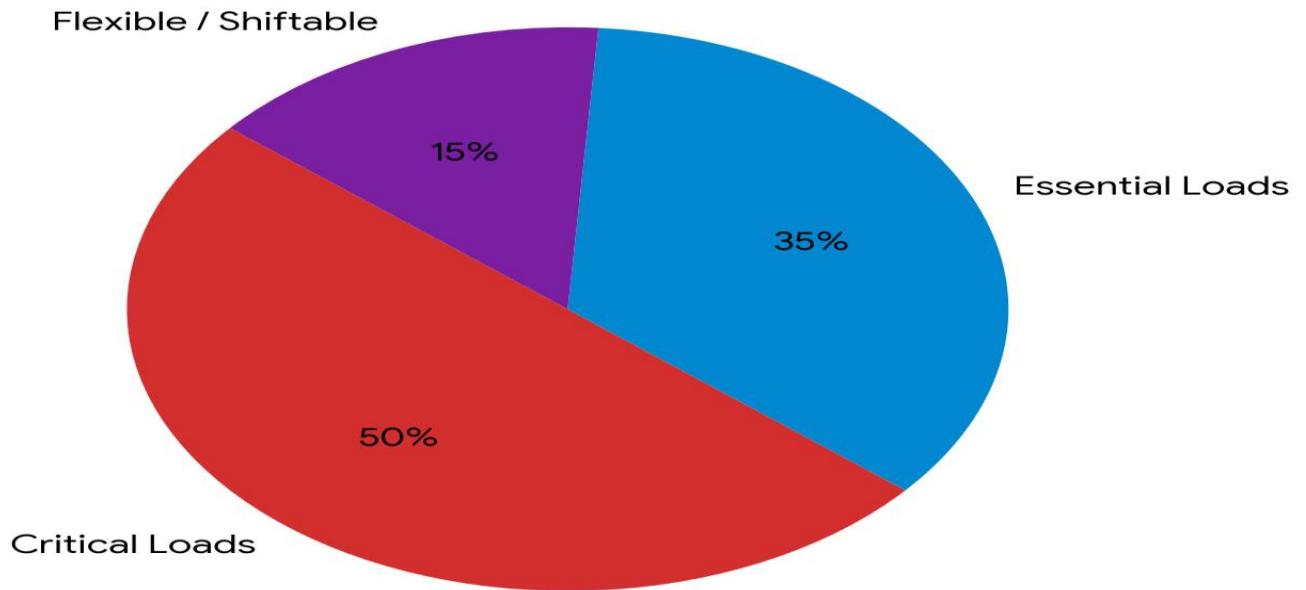
Microgrids must continuously balance energy generation, storage and consumption while accommodating intermittent renewable sources. Conventional management approaches often have limited predictive capability and may react only after a change occurs. This can cause inefficient use of renewable energy, unnecessary battery cycling, peak grid demand and increased operational expense.

The proposed AI-powered platform addresses this problem by monitoring electricity generation and consumption, forecasting electricity demand, optimizing battery storage utilization, balancing electrical loads and presenting real-time operational information.

Objectives

- Monitor energy generation and consumption in real time.
- Forecast electricity demand using AI models.
- Optimize battery charging and discharging decisions.
- Balance electrical loads automatically based on priorities and constraints.
- Integrate renewable energy sources into one coordinated platform.
- Generate energy performance reports and alerts.
- Reduce operational costs and unnecessary grid dependence.
- Improve microgrid reliability and sustainable energy management.

Electrical Load Prioritization Profile



4. Working Principle / System Architecture

The architecture is organized into five logical layers. The sensing layer collects measurements. The communication layer transfers data. The analytics layer cleans data and produces forecasts. The optimization layer selects suitable energy actions. The application layer displays results and reports to operators.

IoT Sensors & Smart Meters	Data Gateway & Cloud/Edge	AI Forecasting Engine	Optimization Controller	Dashboard & Reports
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Step 1: Smart meters measure voltage, current, power, energy and load demand.

Renewable controllers provide generation data, while the battery management system provides state-of-charge and operating status.

Step 2: Data are transmitted through a gateway using secure communication protocols and stored for real-time and historical analysis.

Step 3: The AI engine forecasts demand and, where relevant data are available, renewable generation.

Step 4: The optimizer compares predicted demand, available generation, battery state and load priority to create an operating schedule.

Step 5: Commands or recommendations are sent to controllable loads, storage systems and operators.

5. Key Components and Technologies

Smart Meters and IoT Sensors

These devices collect electrical measurements at generation points, feeders, batteries and loads. They provide the data foundation for monitoring and AI analysis.

Renewable Energy Sources

Solar photovoltaic systems and wind systems can act as distributed energy resources. Their output is monitored so that available renewable energy can be prioritized.

Battery Energy Storage System

The battery stores surplus energy and supplies energy when generation is insufficient or when avoiding expensive peak grid consumption is beneficial.

AI and Machine Learning Engine

Time-series forecasting models can learn daily, weekly and seasonal demand patterns. Model selection depends on available data and project requirements.

Optimization and Load Control

A rule-based or mathematical optimization layer converts forecasts into decisions while respecting battery limits, load priority, power balance and operational constraints.

Cloud/Edge Platform and Dashboard

The platform stores data, runs analytics and displays key performance indicators, alerts and reports.

6. AI-Based Demand Forecasting

Demand forecasting estimates future electricity consumption for a selected time horizon such as the next hour, day or scheduling interval. Input features may include previous demand, time of day, day of week, weather information, occupancy and special operating schedules when such data are available.

Historical Data	Data Cleaning & Features	AI Forecast Model	Predicted Demand	Control Decision
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A typical workflow includes data collection, preprocessing, feature engineering, model training, validation and deployment. Forecast quality can be evaluated using measures such as MAE, RMSE or MAPE.

Possible baseline models include linear regression and moving averages. More advanced implementations may use random forests, gradient boosting, recurrent neural networks or

transformer-based time-series models when the dataset and computing resources justify them.

Engineering Principle

The platform should avoid treating AI predictions as absolute truth. Forecast confidence, data quality and fallback control rules should be considered so that safe operation continues even when data are missing or predictions are inaccurate.

7. Battery Storage Optimization

Battery management is a central part of the proposed system. The controller considers state of charge, available renewable power, predicted demand, grid conditions, battery limits and load priorities.

- Charge the battery when renewable generation exceeds immediate demand and storage capacity is available.
- Discharge when renewable generation is insufficient and stored energy can reduce costly or unnecessary grid use.
- Maintain minimum and maximum state-of-charge limits.
- Avoid excessive switching and unnecessary charge-discharge cycles.
- Reserve energy for critical loads when reliability requirements are high.

Simplified Decision Logic

Generation > Demand	Charge Battery or Flexible Load	Generation < Demand	Battery / Grid Supply	Critical Load	Highest Priority
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The optimization objective can be expressed conceptually as minimizing operating cost and unmet demand while maximizing renewable utilization, subject to power balance and battery constraints.

8. Automatic Load Balancing

Loads can be classified according to importance. Critical loads may include safety, essential computing or process equipment. Flexible loads can be shifted or reduced within agreed operating limits. The load-balancing controller uses this classification together with predicted supply and demand.

- Priority scheduling: supply critical loads first.
- Peak shaving: reduce dependence on the grid during high-demand periods.
- Load shifting: move flexible tasks to periods with higher renewable availability.
- Demand response: adjust controllable loads according to system conditions.
- Fairness and constraints: prevent repeated interruption of the same load and respect user-defined limits.

For a prototype, automatic balancing can be simulated with virtual loads in software. Each load can have a power value, priority level and controllable status. This provides a safe and low-cost demonstration without directly controlling high-voltage equipment.

Expected Benefits

Better matching of generation and consumption can improve reliability, reduce peak demand and increase the useful utilization of locally generated renewable energy.

9. Applications and Real-World Use Cases

- College and university campuses with solar generation and multiple academic buildings.
- Industrial facilities with variable process loads and reliability requirements.

- Residential communities with shared rooftop solar and battery storage.
- Hospitals and essential-service facilities where critical loads require priority supply.
- Remote communities that combine local renewable generation with storage.
- Commercial buildings seeking peak-demand management and energy reporting.

Illustrative Campus Scenario

Consider a campus microgrid with rooftop solar panels, a battery bank and multiple building loads. During high solar production, the platform supplies active demand and stores suitable surplus energy. Before an expected evening demand peak, the forecasting model estimates the required power. The controller can preserve battery capacity for critical loads and shift selected flexible loads to more favorable periods.

The operator dashboard displays current generation, consumption, battery state, predicted demand and recommended actions. This transforms energy data into operational information that can support faster decisions.

10. Advantages and Limitations

Advantages

- Improves visibility through continuous monitoring.
- Supports proactive decisions using demand forecasts.
- Can increase renewable energy utilization.
- Helps reduce peak demand and operating costs.
- Improves prioritization of critical loads.
- Provides historical reports for planning and maintenance.
- Supports scalable integration of IoT, cloud and AI technologies.

Limitations

- AI accuracy depends on sufficient and reliable data.
- Sensor failures and communication delays can affect decisions.
- Battery aging and physical constraints limit optimization.
- Cybersecurity is essential because energy systems use connected devices.
- Initial deployment may require investment in meters, gateways and control equipment.
- Poorly designed automation can create operational risk; validation and human oversight remain important.

Therefore, the proposed system should combine AI prediction with engineering constraints, secure communication, monitoring and fallback rules.

11. Recent Developments and Emerging Trends

Current smart-energy platforms increasingly combine edge computing, cloud analytics, IoT connectivity and AI-based forecasting. Edge processing can reduce response time for local decisions, while cloud platforms support long-term storage and model training.

- Digital twins for simulating microgrid behavior before changing control strategies.
- Explainable AI to help operators understand important factors behind forecasts and recommendations.
- Federated or privacy-aware learning approaches for distributed energy data.
- Reinforcement learning research for adaptive energy scheduling under changing conditions.
- Advanced battery analytics for state-of-health estimation and predictive maintenance.
- Integration of electric vehicles as flexible loads or potential energy resources where supported by infrastructure.

These trends indicate that future microgrids may become increasingly autonomous, but trustworthy design, cybersecurity and transparent human supervision will remain important.

12. Challenges and Future Scope

Challenges

- Intermittent renewable generation and forecasting uncertainty.
- Incomplete, noisy or missing sensor data.
- Interoperability between equipment from different manufacturers.
- Cybersecurity and access control for IoT and control systems.
- Scalability when many meters and loads are connected.
- Balancing cost optimization with reliability and user comfort.
- Validation before deploying automatic control to physical infrastructure.

Future Scope

- More accurate probabilistic forecasting and uncertainty-aware optimization.
- Digital-twin simulation for testing control policies.
- Multi-microgrid coordination and peer energy sharing.
- Integration of electric vehicle charging schedules.
- Predictive maintenance for inverters, batteries and other assets.
- Carbon-aware scheduling based on the environmental impact of energy sources.
- Natural-language interfaces that explain energy performance to operators.

13. Case Study / Prototype Example

A prototype can model a campus microgrid over 24 hours using time-stamped data. Inputs include solar generation, base load, flexible load, battery state of charge and electricity price or a simplified peak/off-peak indicator.

Example Control Cycle

- 1. Read generation and load measurements.
- 2. Store and preprocess incoming data.
- 3. Forecast demand for the next scheduling interval.
- 4. Estimate available renewable generation.
- 5. Check battery limits and reserve requirements.
- 6. Prioritize critical loads.
- 7. Select charge, discharge, grid-import or load-shifting action.
- 8. Update dashboard and record the decision.
- 9. Generate daily or weekly performance reports.

Evaluation Metrics

The prototype can compare an AI-assisted strategy with a simple fixed-rule baseline using renewable utilization, grid energy consumption, peak demand, estimated operating cost, forecast error and percentage of demand served.

This comparison provides a measurable demonstration of whether predictive management improves performance under the selected dataset and simulation assumptions.

14. Conclusion

The AI-Based Smart Microgrid Energy Management and Load Balancing Platform presents an intelligent approach to coordinating renewable generation, battery storage and electricity consumption. By combining IoT monitoring, AI-based demand forecasting and optimization, the system can move from reactive energy management toward predictive and data-driven operation.

The proposed platform addresses key objectives: monitoring generation and consumption, forecasting demand, optimizing storage, balancing loads, integrating renewable sources and generating performance reports. Its expected outcomes include improved energy efficiency, better load balancing, reduced electricity costs, increased renewable utilization, enhanced reliability, sustainable management and data-driven planning.

A practical implementation should begin with simulation or a low-voltage prototype, followed by careful validation of data quality, cybersecurity, control constraints and safety requirements before any real-world deployment.